KARL SUNGMIN MULLER

Senior Software Engineer & Data Scientist — Machine Learning, Computer Vision & Full-Stack Systems

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Professional Summary

PhD-trained software engineer and data scientist with 4+ years shipping production machine learning and computer-vision systems end to end. Builds the full stack — web apps (React, Svelte, TypeScript, Node.js, FastAPI), backend services and data pipelines, and cloud/CI-CD infrastructure (Docker, Kubernetes, Jenkins, AWS/GCP) — alongside the models behind them: CNNs, object tracking, 3D reconstruction, and multi-sensor fusion taken from prototype to deployment. Equally comfortable owning a backend service, a model lifecycle, or a 3D/visualization front-end.

Technical Skills

Programming: Python, JavaScript, TypeScript, SQL, MATLAB, Shell scripting

Software Engineering & Architecture: System design, REST APIs (FastAPI), backend services, distributed data pipelines, test-driven development, Agile

Cloud & Infrastructure: GCP (Cloud Run, Pub/Sub, Cloud Storage, Batch, Managed Instance Groups, Cloud Scheduler, Artifact Registry), AWS, Terraform, Docker, Kubernetes, CI/CD (GitHub Actions, Jenkins, ArgoCD), Linux

Full-Stack & Web: React, Svelte, TypeScript, Node.js, Three.js, Cesium.js, deck.gl, WebGL, Leaflet, Jest

Machine Learning & AI: Deep learning, convolutional neural networks, computer vision, predictive modeling, model deployment, ML performance monitoring (MLOps); agentic coding (Codex, Claude Code, Cursor Agent); LLM-assisted ("autoresearch") model development

Python Libraries: NumPy, scikit-image, scikit-learn, SciPy, PyTorch, XGBoost, HiGHS, OpenCV, pytest, Matplotlib, Shapely, PyVista

Databases: PostgreSQL, PostGIS, pgvector

Geospatial & Remote Sensing: AIS / GPS time-series, H3 hierarchical spatial indexing, GeoParquet; Lidar, RGB, thermal and multispectral data; map and 3D visualization; 3D data-processing algorithms

Computer Vision & 3D: CNNs, RGB and depth cameras, eye tracking, optic flow, object tracking, camera projection, rendering, image processing; 3D engines (Blender, Unity, Open3D), virtual camera rendering

Mathematics: Linear algebra, numerical optimization, graph theory, integer programming

Statistics & Data: Large-scale data analysis, data visualization, statistical modeling

Leadership & Communication: PR review, junior mentorship, pair programming; conference talks and posters; stakeholder meetings; project leadership

Languages: English (professional), Spanish (proficient), German (intermediate)

Experience

Senior Data Scientist — SynMax **Oct 2024 – Present**

Remote

- Built production multi-object vessel tracking using k-partite graph matching, solved with the HiGHS integer-programming solver.
- Tech lead for the rapid integration of two new data sources into the platform, coordinating delivery across teams.
- Built a camera-to-AIS vessel attribution system combining 3D geometric projection with a webcam/AIS caching layer, running GPU inference cheaply on GCP Spot VMs in a managed instance group (MIG).
- Developed machine-learning methods to fuse AIS (GPS vessel time-series) with satellite imagery using XGBoost.
- Implemented cubic-spline smoothing and interpolation to reconstruct clean vessel trajectories from sparse, noisy AIS.
- Created geolocation-correction methods combining OpenStreetMap data with image processing, bringing >50% of images to sub-200 m accuracy (vs. typical kilometer-scale error) and removing the need to reprocess imagery while waiting on delayed correction data.
- Built an automated ML performance-monitoring pipeline: Cloud Scheduler inserts evaluation samples, a custom Cloud Run web UI captures manual annotations into PostgreSQL, and scheduled Cloud Run jobs compute model scores and auto-generate performance reports.
- Led the build-out of an internal package repository on GCP Artifact Registry — now used by 5 teams across 20+ shared packages — streamlining tool sharing and CI/CD.

Engineering Scientist — Applied Research Laboratories **Jul 2023 – Oct 2024**

Austin, TX

- Developed custom 3D visualization tools with React and Cesium, enhancing geospatial data visualization for situational awareness and route planning.

- Built interactive 3D visualization applications with PyVista and VTK, enabling detailed exploration and analysis of complex geometric data.
- Worked with lidar, multispectral, and thermal imaging data to support comprehensive remote-sensing analysis.
- Designed and built front-end and back-end systems for handling geospatial and geometric data.
- Communicated project progress and technical detail to stakeholders, ensuring alignment on project goals.

Senior Software Engineer — EagleView **Apr 2023 – Jul 2023**
Remote

- Partnered with team members to gather requirements and define concrete implementation plans for new features.
- Identified and resolved friction points in internal processes during retrospectives.
- Mentored junior engineers through code review and pair programming, improving team efficiency and knowledge sharing.

Software Engineer II — EagleView **Mar 2022 – Apr 2023**
Remote

- Developed a suite of 3D modeling tools for an in-house application, applying geospatial reference systems, coordinate transformations, image processing, computational geometry, and numerical optimization.
- Played an instrumental role in launching a new 3D modeling app by reverse-engineering poorly documented legacy code to integrate new tooling.
- Won the company hackathon with an innovative approach to modeling 3D roof geometry; the tool accelerated operator efficiency by up to 20x.
- Implemented test-driven development and CI/CD with Jenkins, ArgoCD, and EKS to ensure software quality and efficiency.

Graduate Research Assistant / Postdoctoral Fellow **Mar 2017 – Mar 2022**
Vision, Cognition, and Action VR Lab, UT Austin

- Built a novel complex-terrain navigation dataset using the photogrammetry package Meshroom combined with eye-tracking and IMU data.
- Developed a processing pipeline deployed on UT's TACC supercomputing system to process large volumes of video data efficiently.
- Applied CNNs and boosting algorithms to analyze and model complex terrain navigation in humans.
- Supervised undergraduates and coordinated tasks across the project.

Graduate Research Assistant **Jan 2018 – Jul 2018**
Huth Lab, UT Austin

- Reconstructed 3D hand pose from multiple 2D views using open-source pose-estimation methods.
- Estimated 3D object pose from a 2D projection silhouette and a 3D model.

Education

PhD, Neuroscience — The University of Texas at Austin, 2021. Thesis: *Modelling Visually Guided Natural Locomotion*.

BS, Neuroscience (Certificate in Computational Science and Engineering) — The University of Texas at Austin, 2016.

Awards

NIH Big Data to Knowledge (BD2K) Fellowship, 2018–2020.

Selected Publications

Muller, K., Bonnen, K., Shields, S. M., Panfili, D., Matthis, J. S., & Hayhoe, M. (2024). Analysis of foothold selection during locomotion using terrain reconstruction. *eLife*, 12, RP91243.

Muller, K., Matthis, J., Bonnen, K., Cormack, L. K., Huk, A. C., & Hayhoe, M. (2023). Retinal motion statistics during natural locomotion. *eLife*, 12, e82410.

Panfili, D. P., Muller, K., Bonnen, K., & Hayhoe, M. M. (2026). Visual control of walking using terrain reconstructions. *Scientific Reports*.

Zhang, R., Liu, Z., Zhang, L., Whritner, J. A., Muller, K. S., Hayhoe, M. M., & Ballard, D. H. (2018). AGIL: learning attention from human for visuomotor tasks. In *Proceedings of the European Conference on Computer Vision (ECCV)*, 663–679.

Full publication list (10 papers) and conference presentations available on request.